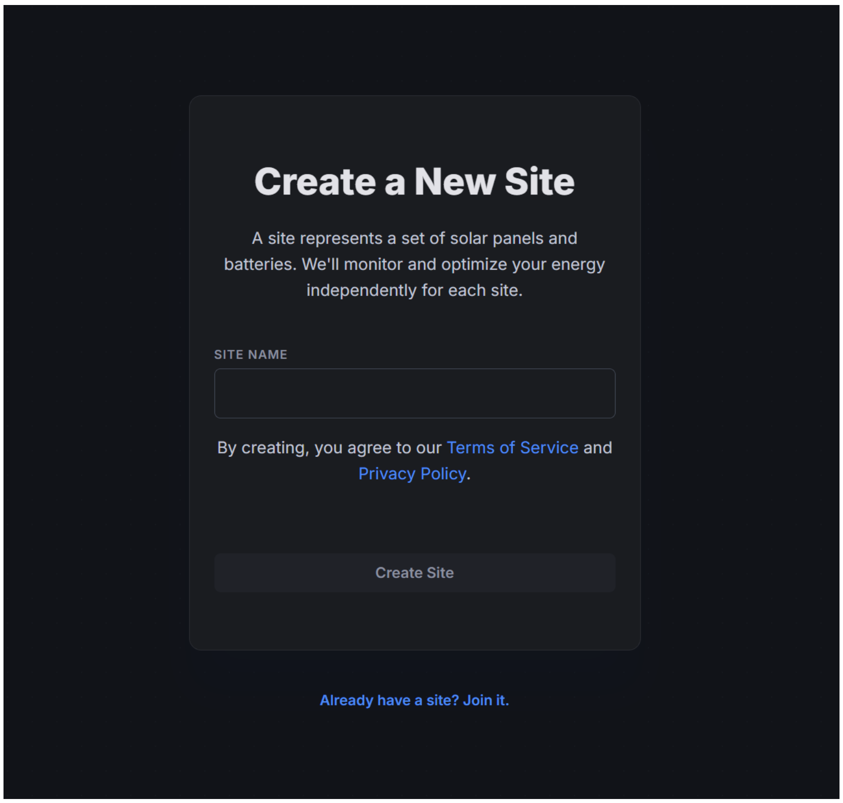
RateRudder System User Guide

Quick Start & Step-by-Step Walkthrough

This guide walks you through setting up and using the RateRudder optimization system. Learn how to create your site, connect your devices, customize grid restrictions, configure location parameters, monitor real-time savings automation, and read the 24-hour simulation forecasts.

Step 1: Creating a New Site

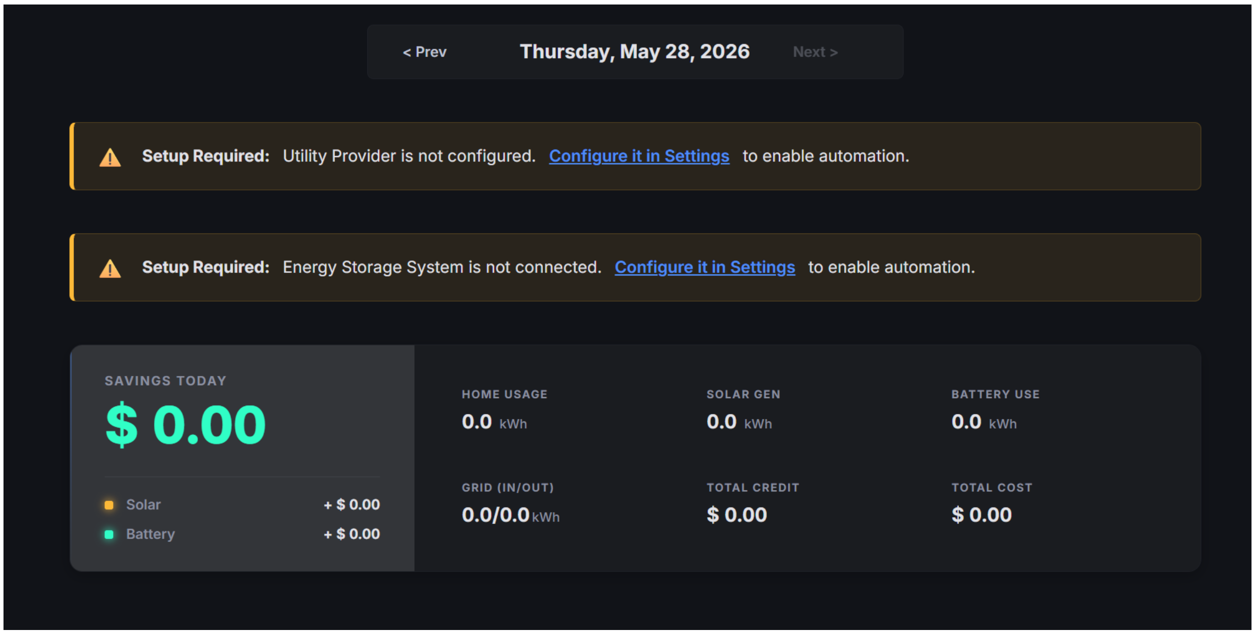
To begin using the RateRudder optimizer, your first step is to initialize a Site. A Site serves as a workspace representing your physical installation of solar panels, batteries, and meter controllers. By monitoring and optimizing each site independently, the system delivers personalized charging and discharging schedules based on localized factors.



NOTE: A site can also be simplified as a residence. In the rare case your residence contains multiple battery controllers or solar arrays, you might need to create multiple distinct sites to manage them independently.

Step 2: Initial Dashboard & Setup Warnings

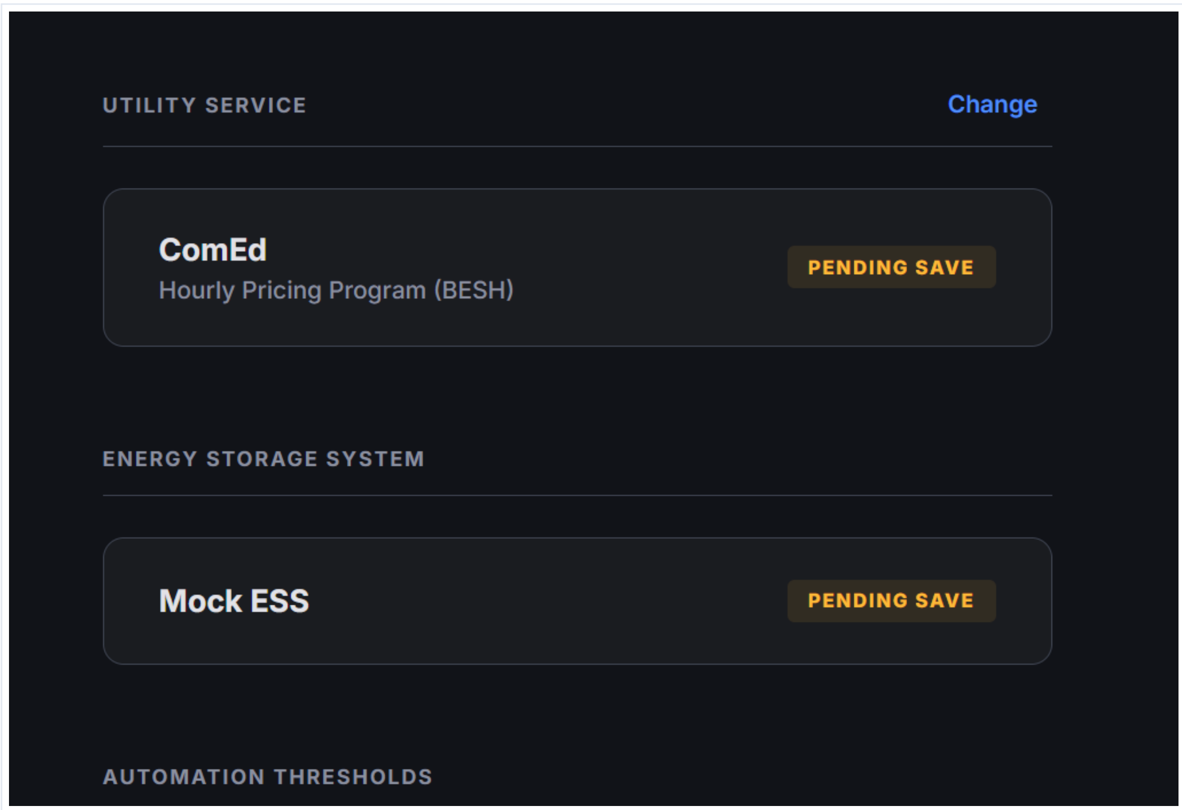
When a site is first created, you will be taken to the main dashboard. Because no hardware or pricing configurations have been established, you will see critical warning banners at the top of the page. These banners indicate that the system cannot automate your energy storage until setup is completed.



IMPORTANT: Before anything will work, you will need to configure your battery system and your utility provider information. Follow the links in the warning banners or navigate to the Settings tab to configure these requirements.

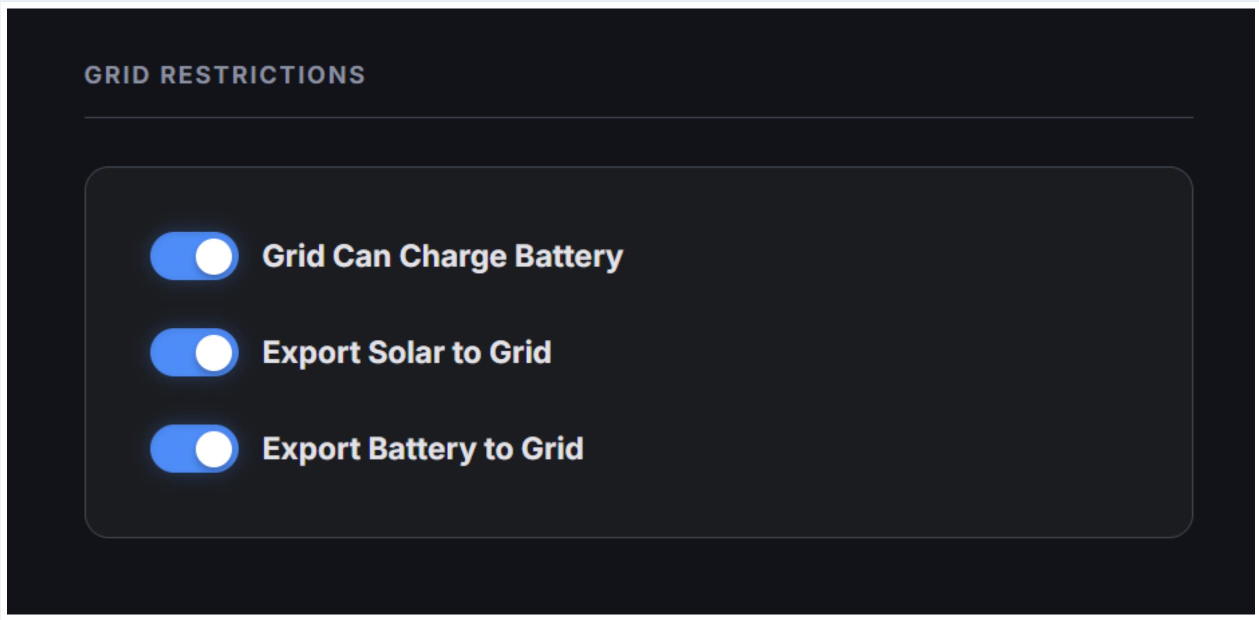
Step 3: Utility & Energy Storage System Configuration

Navigate to the device configuration section in Settings to connect your household components. First, select your electricity Utility Service to enable real-time price tracking. Next, select your Energy Storage System (ESS) type to establish hardware communication. Once selected, they will display a 'PENDING SAVE' badge until you commit the changes.



Step 4: Grid Restrictions Settings

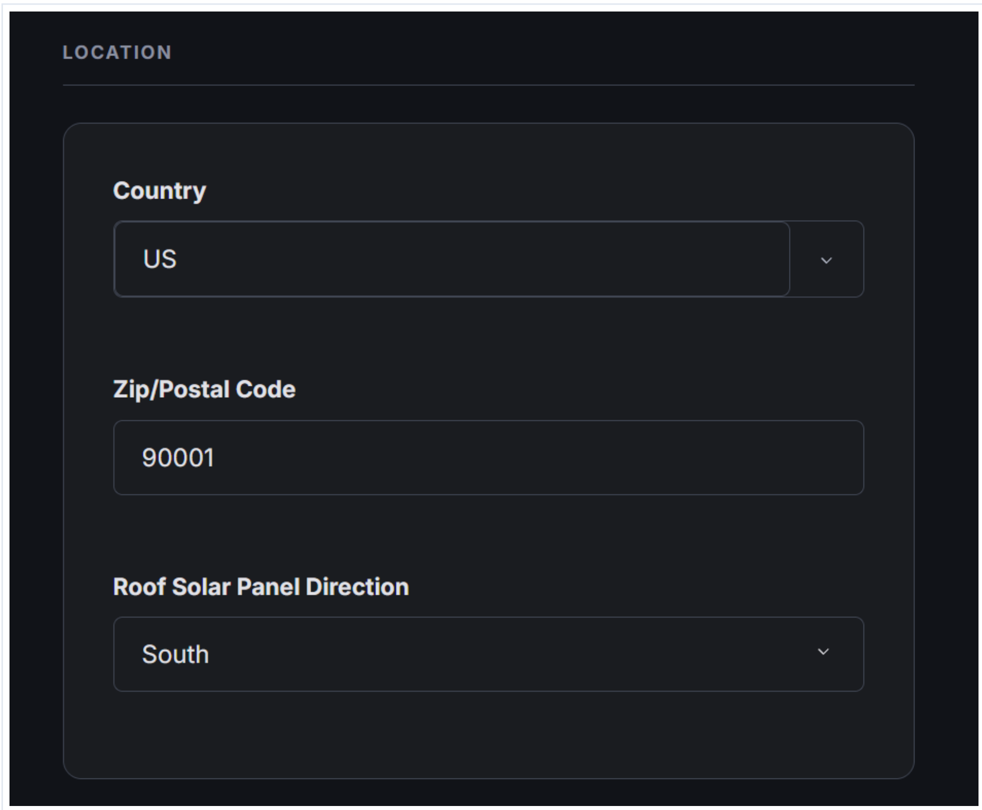
The Grid Restrictions card governs how your battery and solar system interact with the local utility grid. Using the toggle switches, you can dictate permissions such as allowing the grid to charge the battery (useful for off-peak rate charging) and allowing solar energy or battery discharges to be exported back to the grid for credits.



IMPORTANT: Make sure to choose the grid restrictions that are applicable to your installation. Selecting unsupported behaviors can lead to billing errors or compliance issues with your utility provider.

Step 5: Location Settings

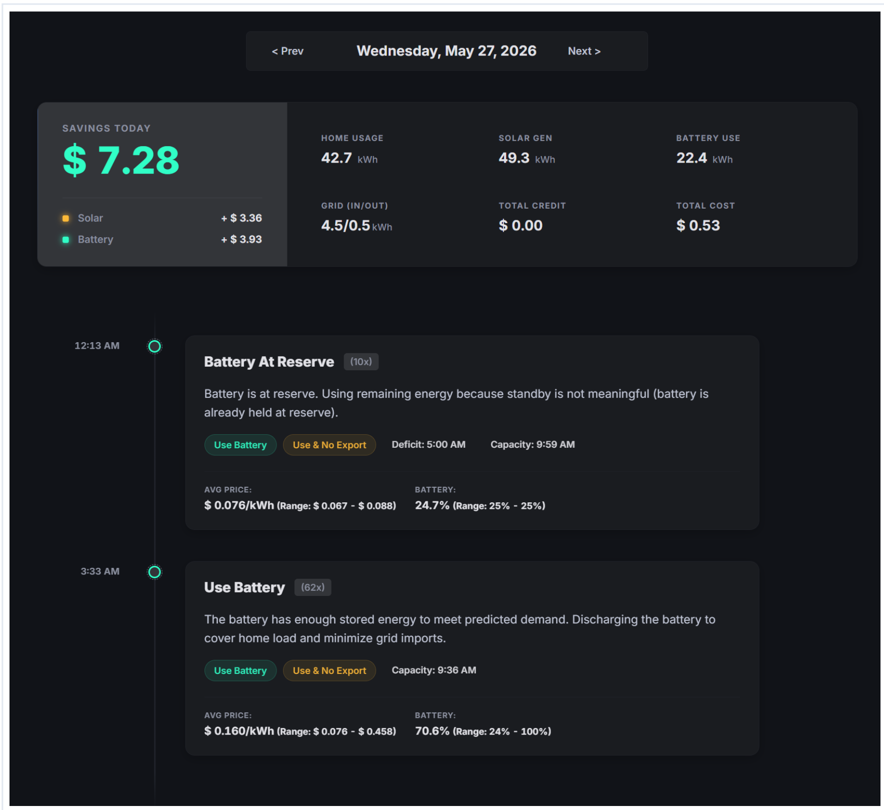
Accurate location coordinates are critical for solar generation predictions. By specifying your Country and Zip/Postal Code, the optimizer retrieves regional weather forecasts. Providing your Roof Solar Panel Direction (e.g., South, West) allows the system to model the sun's angle and calculate expected hourly solar generation with high precision.



PRO-TIP: A south-facing solar panel direction is typical in the Northern Hemisphere for maximum total yield, but make sure to enter your actual physical panel orientation to keep predictions accurate.

Step 6: Dashboard & Active Automation Monitoring

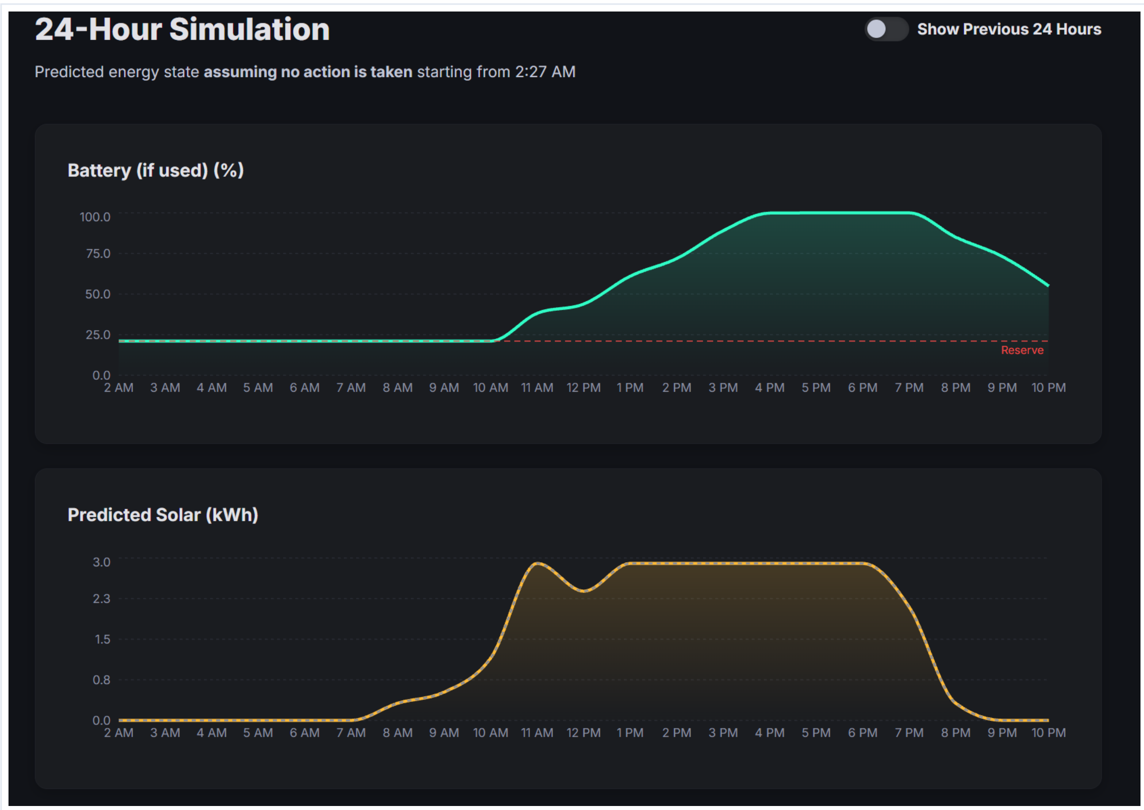
Once configuration is saved, the dashboard begins displaying active energy data and savings statistics. The top banner tracks your 'Savings Today' (divided into solar and battery contributions), home usage, solar generation, battery usage, and net grid costs/credits. Below, the system logs a chronological timeline showing exactly when and why the optimizer chose automated actions (e.g. discharging at high rates).



PRO-TIP: The timeline events provide full transparency. For example, you can see the exact time the battery reached reserve capacity or when the system triggered a discharge to minimize grid imports during peak price hours.

Step 7: 24-Hour Simulation & Forecasting

The 24-Hour Simulation charts project your energy state over the next day assuming no manual overrides. The top graph details the battery state of charge (%) relative to your Reserve threshold (e.g., 25%). The bottom graph maps the predicted solar generation (kWh) hour-by-hour based on weather forecasts and location parameters, allowing you to anticipate battery charging cycles.



NOTE: The forecast page shows an estimated look at the next 24 hours. Because it relies on predictive models for weather and household consumption, actual battery levels may vary slightly as the day unfolds.